Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Early-career software engineer with 2+ years delivering **Python/C++ systems**, **event-driven services**, and **test-focused engineering workflows** for latency-sensitive, high-throughput production environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Built **production Python services** and data interfaces for 500+ users, reducing query latency from **1.2s to 350ms** through structured optimization and profiling.
- Implemented **failure-aware orchestration controls** for 5K+ monthly jobs, strengthening system reliability and improving incident response in distributed execution workflows.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Developed **full-stack product features** for 10K+ daily requests using React/Flask APIs, shipping maintainable modules through incremental Agile delivery and code reviews.
- Improved **CI/CD and automated validation** to reduce release turnaround by **60%**, preserving deployment quality in fast-moving product iteration cycles.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Built **benchmarking pipelines** over 100K+ trajectories, enabling repeatable experiment execution and quantitative comparison for model and system behavior analysis.
- Authored **technical summaries and debugging diagnostics** that translated research outcomes into engineering tasks, improving cross-functional planning and execution quality.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **event-driven API/ETL workflows** with Python/Java, AWS Glue, and S3, improving response times by **30%** across distributed enterprise processing systems.
- Maintained **20+ CI-integrated test suites** with Jenkins, PyTest, and Mockito, increasing pipeline efficiency by **68%** and reducing regression defects in production.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Implemented **MERN + WebSocket services** for eight workflows and 2M+ synchronized records, improving consistency and responsiveness for concurrent user operations.
- Automated **containerized deployments** through Docker/Jenkins pipelines, cutting rollout time by **50%** and standardizing release behavior across staging/production paths.

Skills

Languages: **Python**, **C++**, Java, JavaScript/TypeScript, SQL, Bash
Systems: event-driven architectures, distributed services, multithreaded fundamentals, Linux/UNIX
Engineering: test-driven development, refactoring, code reviews, debugging, quantitative performance analysis
Build & Infra: Docker, Jenkins, GitHub Actions, Make/CMake, CI/CD, PyTest, Mockito
Collaboration: Agile/Scrum, mentoring feedback loops, technical documentation, cross-functional execution

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, and MLFQ** scheduling with runtime metrics and visualization, reducing debugging time by **20%** via deterministic simulation traces.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Developed a **containerized data-processing API** on AWS ECS/Fargate with automated validation, improving uptime and operational reliability for production-scale workloads.

Market Event Stream Analyzer | *C++, Python, Kafka, PostgreSQL*

- Built a **low-latency event ingestion and scoring pipeline** with statistical alerts and replay tooling, enabling faster anomaly detection and strategy signal inspection.